

Master thesis on Sound and Music Computing
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Modelling Timbre for Neural Violin Synthesis

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Abstract

This thesis investigates timbre-informed neural violin synthesis. Existing neural approaches such as DDSP rely mainly on fundamental frequency and loudness, or on spectrogram-based features that are high-dimensional and difficult to interpret. As a compact alternative, we propose a conditioning tuple of fundamental frequency (F0), energy, a harmonic-magnitude embedding, and an inharmonicity feature. The harmonic embedding captures stable harmonic structure, while the inharmonicity feature quantifies deviations that shape violin timbre.

A DDSP-based decoder was conditioned on this tuple and evaluated on the Violin Etudes dataset with CHR-F0 and SWIPE-F0 trajectories. Results show that adding harmonic and inharmonic features improves synthesis compared with a baseline DDSP, with the CHR-F0 configuration achieving the most balanced outcome across spectral and perceptual metrics.

The contributions are threefold: (i) a compact, interpretable timbre representation for violin synthesis, (ii) experimental results showing that this representation improves synthesis quality, and (iii) open resources including code, checkpoints, and audio examples. These findings suggest that lightweight timbre control enhances both quality and controllability in violin synthesis, providing a basis for future perceptual evaluation and physically informed models. The main repository containing code and audio examples is available at:

github.com/kenzhuti/timbre-informed-ddsp-violin.

Keywords: DDSP; violin timbre; harmonic-magnitude embedding; inharmonicity; neural audio synthesis

Chapter 1

Introduction

1.1 Motivation

Timbre is a multidimensional attribute, distinct from pitch and loudness, shaped by acoustic, perceptual, and cognitive factors [1, 2]. The violin, with its fretless design and wide range of articulations such as vibrato, portamento, and varied bowing techniques, exhibits particularly rich timbral variation [3]. Capturing the nuances of violin performance in synthesis remains challenging, as models still struggle to achieve natural and realistic audio. Prior studies have highlighted limitations such as the subjectivity of expressive annotations and poor generalisation on scarce data [4], the production of unrealistic timbre despite improvements in timing and dynamics [5], and artefacts or the loss of timbral continuity between notes [6].

DDSP-based neural synthesis frameworks highlight this gap. **Differentiable Digital Signal Processing (DDSP)** conditions on F0 and loudness, with an optional latent $z(t)$ [7]. While compact, $z(t)$ behaves as a black-box, which reduces interpretability and risks overfitting on small datasets. Moreover, only F0 plus loudness is considered less informative. By contrast, **the source-filter GLOttal-flow LPC Filter with sample-wise LPC (GOLF-ss)**, trained as well as synthesised with sample-wise LPC) employs log-F0 and Mel-spectrograms, which are more informative but computationally heavy and less interpretable [8].

Thus, current approaches trade off compactness against informativeness. For violin synthesis, where nuanced control is essential, there is a need for conditioning features that remain compact and interpretable while providing richer timbral information. This motivates the objectives defined in Section 1.2.

1.2 Objectives

This thesis pursues three interrelated objectives:

- **Timbre Modelling for Neural Synthesis** Represent violin timbre in a compact, interpretable, and multifaceted manner by focusing on two complementary aspects:
 - Harmonic dimension — captured through harmonic magnitude clustering and embedding.
 - Inharmonic dimension — addressed via explicit inharmonicity measures.

These components serve as meaningful conditioning features for neural audio synthesisers.

- **Compact and Interpretable Conditioning Representations** Design input representations for neural synthesisers that balance compactness, interpretability, and informativeness, offering a more efficient and explainable alternative to dense spectrogram-based input.
- **Synthesis Quality under Timbre-Conditioned Systems**
 - Evaluate whether timbre-informed conditioning can achieve satisfactory synthesis and reconstruction quality.
 - Investigate the trade-off between lightweight feature-based conditioning and spectrogram-based conditioning approaches such as GOLF-ss.

Together, these objectives aim to advance controllable and interpretable timbre-informed neural violin synthesis.

1.3 Structure of the Report

The remainder of this thesis is structured as follows. Chapter 2 reviews the literature on timbre modelling, from traditional descriptors to modern computational embeddings, surveys neural audio synthesis with emphasis on DDSP and violin, and finally datasets for violin timbre research. Chapter 3 introduces the proposed *Timbre-Informed DDSP Violin* architecture, describing timbre feature extraction (hmag embedding, inharmonicity), CHR-based F0 estimation, and decoder design. Chapter 4 reports experiments on Violin Etudes: a player-acoustics pilot, timbre modelling via hmag clustering/embedding, **ablations across conditioning schemes** (baseline, +hmag, +inharm, full tuple) under SWIPE- and CHR-F0, a hybrid architecture with mel-spectrogram concatenation, objective evaluations (MCD, MSS, PESQ, FAD), and a controlled single-player study. Chapter 5 discusses contributions, limitations, and directions for perceptual evaluation and physically informed synthesis. Finally, Chapter 6 concludes by summarising findings and emphasising the value of compact timbre conditioning for controllable violin synthesis.

Chapter 2

State of the Art

2.1 Timbre Modelling and Representation

2.1.1 Traditional Approaches

Early attempts to model timbre adopted a compositional perspective, separating its attributes into temporal, spectral, and time–frequency domains.

In the **temporal domain**, the widely used attack–decay–sustain–release (ADSR) envelope provides a coarse but effective abstraction of sound evolution. The duration and relative balance of these phases are crucial determinants of instrument identity and perceptual sharpness. For instance, short attack times typically characterise percussive or plucked timbres, whereas sustained envelopes are more common in bowed or wind instruments.

In the **spectral domain**, descriptors such as spectral centroid, spectral flux, formant frequencies, and harmonic-to-noise ratios have been proposed as correlates of perceived brightness, stability, or noisiness. These measures allow for compact characterisations of sound colour and are widely used in music information retrieval tasks. Nevertheless, their interpretability is often limited, since multiple acoustic cues may contribute jointly to perceptual judgements.

The **time–frequency domain** provides a richer representation by mapping spectral energy across time. Spectrograms and derived transforms such as the constant-Q transform (CQT), which uses geometrically spaced frequency bins and is well-suited for pitch and music analysis [9] or wavelets provide multiresolution analysis with good time resolution at high frequencies and good frequency resolution at low frequencies [10], capturing transient phenomena, inharmonicity, and vibrato. These representations are standard in computational musicology and serve as the basis for higher-level descriptors such as mel-frequency cepstral coefficients (MFCCs), which have become a cornerstone of speech and music analysis [11].

Beyond acoustic decomposition, **psychological and perceptual methods** have provided fundamental insights into timbre. One of the most influential paradigms is the use of **similarity ratings**, in which listeners compare pairs of sounds and rate their perceptual similarity. The resulting dissimilarity matrices are subjected to multidimensional scaling (MDS), yielding low-dimensional *timbre spaces* in which proximity reflects perceptual similarity [1]. These spaces often reveal two or three salient perceptual dimensions, such as spectral centroid (brightness), spectral flux (roughness), or temporal envelope, which correspond closely to acoustic descriptors.

In addition to similarity ratings, the **semantic and textual approaches** study how musicians and listeners describe the timbre verbally. Analyses of metaphorical descriptors (e.g., *bright*, *warm*, *rough*) and cross-modal correspondences (e.g., linking timbre to colour or touch) have led to the construction of *semantic timbre spaces* [2]. Such approaches highlight cultural and linguistic influences on timbre perception and offer valuable perspectives for bridging computational and perceptual models.

Taken together, these traditional methods provide both signal-based and perception-based frameworks for timbre representation. They remain influential as baselines for modern approaches, despite their limitations in capturing higher-order interactions between pitch, articulation, and expressive nuance.

2.1.2 Advanced Computational Approaches

While traditional and psychological approaches provide foundational insights, more recent work has shifted towards computational frameworks that learn timbre representations directly from data. These methods leverage clustering, disentanglement, hierarchical embeddings, and neural codecs to capture the complexity of timbre in scalable and interpretable ways.

Clustering methods represent an early data-driven approach to timbre modelling. Pérez and Ramírez [12] applied k-means clustering to violin recordings, partitioning timbral features into 32 distinct clusters. This work highlighted how cluster-based models can capture subtle timbral variations within a single instrument, offering a more nuanced representation than handcrafted descriptors. Extending this idea to cross-modal associations, Mannone et al. [13] investigated links between orchestral timbres and colour perception. By clustering both timbre features and colour labels, they demonstrated systematic correspondences between auditory and visual modalities, suggesting that timbre spaces may map naturally onto multimodal perceptual categories.

Pitch–timbre disentanglement has become a central goal for neural audio modelling. Luo et al. proposed Gaussian mixture variational autoencoders (GMVAE) to learn disentangled pitch and timbre embeddings for instrument sounds [14], ensuring that pitch-invariant timbre features can be extracted and manipulated. Building on this, Luo et al. further demonstrated source-level disentanglement in multi-instrument music, enabling separate control of pitch and timbre for individual sources [15]. Similarly, Tanaka et al. [16] introduced a VAE-based metric learning approach that enforces separation between pitch and timbre in latent space, supporting robust retrieval and manipulation tasks. Together, these approaches reflect a broader trend of integrating disentanglement objectives into generative models, allowing timbre to be manipulated independently from pitch while maintaining musical coherence. In parallel, **the original DDSP** framework conditions synthesis on F0 and loudness, with an optional Mel-frequency cepstral coefficients (**MFCCs**)-

based time-varying **latent** $z(t)$ that encodes residual timbral information (e.g., spectral envelope) [7], which provides a flexible extension for modelling aspects of timbre beyond pitch and loudness.

Beyond disentanglement, **hierarchical timbre embeddings** have been explored using non-Euclidean latent spaces. Nakashima et al. [17] proposed hyperbolic embeddings for musical instrument sounds, capturing hierarchical relations between timbres (e.g., instrument families and subtypes). By representing timbre in hyperbolic rather than Euclidean space, their approach encodes parent–child relationships more naturally, aligning with how humans conceptualise instrument taxonomies.

Finally, **neural audio codecs** have introduced a new class of timbre representations based on discrete acoustic tokens. LSCodec [18] achieves low-bitrate and speaker-decoupled encoding by training a VAE with speaker perturbation to form a bottleneck, quantising the latent space with a single VQ codebook, and reconstructing with a context-aware vocoder conditioned on WavLM embeddings. Crucially, LSCodec removes redundant timbre information at the bottleneck stage and re-injects controllable timbre during synthesis, offering both efficiency and controllability. Although developed in the speech domain, such codecs hold strong potential for music applications, including violin timbre modelling, where disentangling performer identity from acoustic colour is essential.

Taken together, these advanced approaches extend timbre modelling beyond handcrafted descriptors towards flexible, interpretable, and controllable computational representations. They pave the way for integrating timbre embeddings directly into expressive synthesis frameworks, a theme further developed in subsequent sections.

2.1.3 Summary

Timbre modelling has progressed from **handcrafted acoustic descriptors** to **perceptual frameworks**, and more recently to **computational embeddings**. Traditional methods (e.g., ADSR, spectral centroid, spectrograms) offer basic signal-level characterisation, while psychological approaches such as **timbre spaces** and **se-**

mantic analyses link acoustic features with human perception [1, 2].

Contemporary computational approaches extend these foundations. Clustering captures latent timbral categories [12, 13], disentanglement methods separate pitch from timbre [14, 15, 16], DDSP models optionally timbre via an optional residual latent z [7], and hyperbolic embeddings model hierarchical relations [17]. Neural codecs such as **LSCodec** further compress and decouple timbre into discrete tokens for controllable synthesis [18].

Despite their progress, these approaches face limitations: traditional models are coarse, perceptual models depend on subjective variability, and data-driven methods often lack interpretability. These challenges motivate the search for unified, perceptually grounded, and more interpretable and more effective timbre embeddings suitable for expressive violin synthesis.

2.2 Audio Synthesis for Music

This section focuses on audio synthesis methods with an emphasis on music, while also introducing a few of approaches developed for speech and sound effects that are technically transferable and offer insights for music and violin synthesis.

2.2.1 Taxonomy

Figure 1 presents an overview of audio synthesis taxonomy based on Hayes et al. [19], which itself builds on earlier classifications by Schwarz [20] and Bilbao [21]. Broadly, synthesis methods can be divided into **parametric** and **data-driven** families.

On the **parametric** side, physical models (e.g., digital waveguides, finite-difference methods) aim to replicate the physics of sound production, while signal models (e.g., additive, subtractive, and source-filter synthesis) describe sound as parameterised transformations of signal representations. These methods are interpretable and compact but often limited in capturing expressive variation.

On the **data-driven** side, concatenative approaches construct new audio by recom-

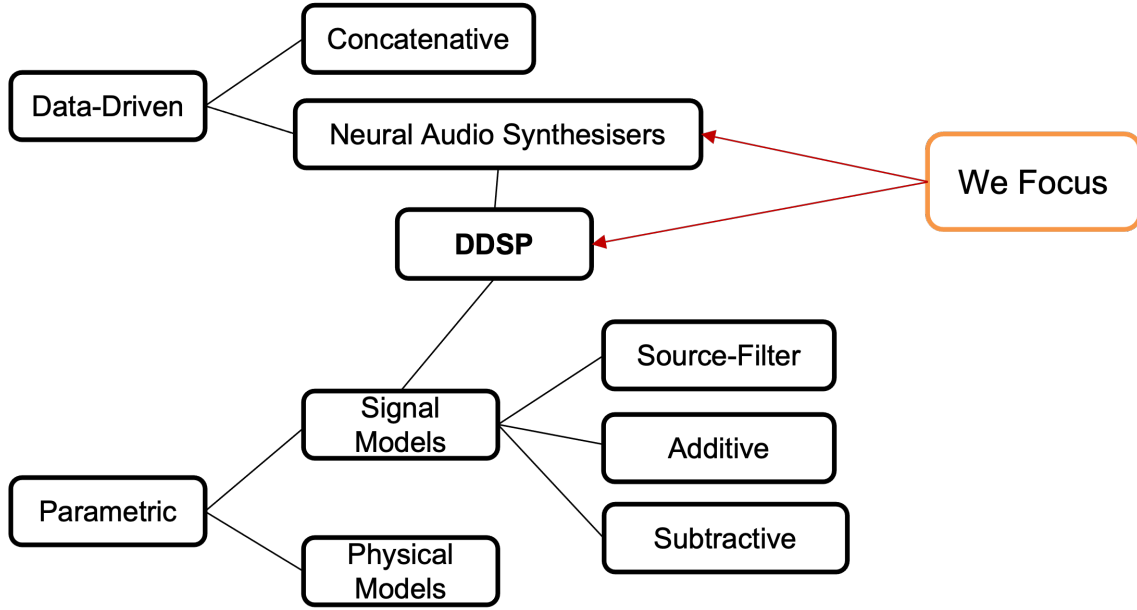


Figure 1: Audio synthesis taxonomy based on Hayes et al. [19].

binning samples from a pre-recorded corpus, whereas neural audio synthesisers learn to generate audio directly from data. The latter include autoregressive models (e.g., WaveNet), flow-based models, GAN-based vocoders, diffusion models, and hybrid systems.

Differentiable Digital Signal Processing (DDSP) lies at the intersection of these two paradigms. By implementing classical DSP operations differentiably, DDSP enables loss gradients to be backpropagated through interpretable signal models, thus combining the **controllability of parametric methods** with the **expressive power of neural networks**. As Hayes et al. argue, this inductive bias addresses some of the robustness and interpretability limitations of purely data-driven models [19].

In this thesis, we focus specifically on **Neural Audio Synthesisers** and, within them, **DDSP-based models**. Section 2.2.2 will review key DDSP approaches, while Section 2.2.3 will discuss other neural models developed for violin synthesis. presents an overview of **audio synthesis taxonomy** based on Hayes et al. [19], which itself builds on earlier classifications by Schwarz [20] and Bilbao [21]. Broadly, synthesis methods can be divided into **parametric** and **data-driven** families.

2.2.2 DDSP-based

Differentiable Digital Signal Processing (DDSP) integrates interpretable signal models with neural networks by allowing gradients to backpropagate through classical DSP modules. This provides domain-specific inductive biases, efficient training, and improved controllability compared to purely black-box models [19]. Since its introduction [7], a growing body of research has applied DDSP to diverse audio synthesis tasks, from speech and singing voice to musical instruments and sound effects. In this subsection, we review representative works in chronological order, highlighting methodological advances and their relevance to music and violin synthesis.

2025. Yeh et al. introduced *DDSP Guitar Amp*, an interpretable model for guitar amplifier emulation. By modularizing the preamp, tone stack, power amp, and output transformer as differentiable components, their approach combined physical interpretability with computational efficiency, achieving real-time performance with less than 10% of the operations required by black-box neural baselines [22].

2024. Yu and Fazekas proposed *Differentiable Time-Varying Linear Prediction* (GOLF-ss, trained as well as synthesised with sample-wise LPC), generalising their earlier GOLF vocoder (for singing voice and speech) [23] to handle sample-wise time-varying coefficients for improved analysis-by-synthesis [8]. Liu et al. presented *DDSP-SFX*, a framework for acoustically guided sound-effects generation, incorporating transient modelling and frame-level timbre control [24]. Gutiérrez explored *Statistics-Driven Texture Sound Synthesis* using DDSP-based architectures, extending DDSP beyond pitched instruments to complex noisy textures. His thesis introduced novel processors (SubEnv Synth, P-VAE Synth) and a TextStat loss, targeting the statistical properties of texture sounds [25].

2023. Yu and Fazekas developed *GOLF*, a singing voice synthesiser that employs glottal-flow inspired wavetables and a differentiable LPC filter. This interpretable design allowed efficient phase modelling and demonstrated competitive performance with significantly reduced parameters and faster inference compared to black-box

vocoders [23]. Renault et al. extended DDSP to polyphonic piano synthesis in *DDSP-Piano*. Their model incorporated explicit inharmonicity modelling and MIDI conditioning, producing realistic piano timbres while retaining interpretability [26]. Li et al. proposed *SnakeGAN*, a universal GAN-based vocoder that integrates DDSP priors as coarse-grained signals, combined with periodic nonlinearities via the Snake activation function. This hybrid improved robustness to out-of-domain tasks, such as unseen instruments or singing voices [27].

2022. Wu et al. introduced *MIDI-DDSP*, a hierarchical framework bridging symbolic MIDI control with DDSP synthesis parameters. Their three-level hierarchy (notes, performance, synthesis) enabled both high realism and detailed expressive control, demonstrated across multiple instruments [28]. Caspe et al. presented *DDX7*, a differentiable FM synthesiser inspired by the Yamaha DX7. By constraining FM synthesis within a differentiable architecture, they enabled continuous control and audio-driven resynthesis of complex timbres [29]. Kawamura et al. proposed a *DDSP Mixture Model* for extracting synthesis parameters from mixtures of harmonic sounds, avoiding explicit source separation by fitting additive DDSP autoencoders directly to mixtures [30]. Finally, Shan et al. developed a *Differentiable Wavetable Synthesiser*, which learns dictionaries of periodic waveforms end-to-end, enabling efficient real-time synthesis and one-shot learning from short audio examples [31].

Summary. The progression of DDSP research illustrates a trend from monophonic harmonic models toward increasingly general, interpretable, and controllable synthesis frameworks. Early works demonstrated feasibility and efficiency, while recent efforts target polyphony, time-varying filters, hybrid GAN integration, and interpretable modelling of complex systems. These developments pave the way for violin-focused applications, where explicit modelling of harmonic structure, inharmonicity, and expressive performance is essential.

2.2.3 Neural Audio Synthesisers for Violin

This subsection reviews neural models specifically developed for violin sound synthesis.

Diffusion-based Models. Kim et al. introduced *ViolinDiff*, a two-stage diffusion framework that explicitly estimates MIDI pitch-bend curves and incorporates them into mel-spectrogram synthesis. This conditioning improves the modelling of vibrato and pitch nuances, resulting in more expressive and realistic violin audio [32].

Comparative Expressive Synthesis. Hung et al. conducted a comparative study of neural architectures for expressive violin performance synthesis. They benchmarked multiple approaches, including end-to-end and parameter-controlled pipelines, with respect to dynamics, articulation, and expressive control, providing a valuable reference for future research [4].

Efficient Neural Pipelines. Volhejn proposed lightweight DDSP-CNN architectures for violin timbre transfer. His models (DDSP-CNN, DDSP-CNN-Tiny) demonstrated that even extremely small models with as few as 2.5k parameters can convert arbitrary input audio into violin-like output, highlighting efficiency gains for real-time or embedded applications [33].

Hybrid Neural–Physical Models. Nizami and Lestari presented a *DT-Neural Parametric Violin Synthesizer*, combining decision-tree-based performance control with neural prediction of bowing parameters and a digital waveguide synthesiser. This hybrid approach preserved physical interpretability while enhancing timbral realism [5]. Jonason proposed the *control-synthesis* approach, which separates a control model (predicting intermediate features such as pitch and loudness) from a synthesis model (a DDSP-based decoder). Demonstrated on violin and flute, the system enabled expressive, MIDI-controllable synthesis from unannotated audio recordings [6].

2.3 Datasets for Violin Timbre Research

Violin-focused audio datasets remain relatively scarce. Table 1 summarises the main available datasets, while below we provide short descriptions of their scope, annotations, and notable features.

Table 1: Violin sound datasets in reverse chronological order.

Dataset	Author (Year)	Mono/ Poly	Size	Annotations	Violinist #	Recording Condition
MUSC [34]	Tamer et al. (2023)	Mono	34 h	High-resolution F0 (5.8 ms / 10-cent) & note transcription; vibrato & portamento; audio-score alignment	22	Various (YouTube curated)
Solo Dataset [35]	Zhao (2022)	Mono	11 h	Onset (partial: 5 h)	22	Studio
Concerto Dataset [35]	Zhao (2022)	Poly	18 h	Vibrato; onset (partial: 3924×9 notes)	9	Various
Violin Etudes [36]	Tamer et al. (2022)	Mono	27.8 h	Frame-level F0 labelling	21	Various (YouTube curated)
Bach Violin Dataset [37]	Dong et al. (2022)	Mono	6.5 h	Audio-score alignment	17	Various
OrchideaSOL (subset) [38]	Cella et al. (2020)	Mono	15k files for all instruments	Extended techniques (e.g., <i>sul ponticello</i> , <i>col legno</i>)	Multi	Studio
URMP [39]	Li et al. (2019)	Poly	11 pieces	Audio-score alignment; F0; onset; video	Multi	Studio
RWC-MDB-I-2001(subset) [40]	Goto et al. (2002)	Mono	57 files	Pitch; dynamic level; playing style (e.g., sustain, staccato, tremolo); instrument ID; manufacturer ID	4	Studio

Violin Etudes. The Violin Etudes dataset contains 27.8 hours of monophonic recordings from 21 violinists, collected mainly from curated YouTube material. Its key contribution is frame-level F0 annotation, making it a reliable benchmark for pitch-tracking and performance analysis. While limited to solo monophonic material, the dataset covers diverse playing conditions and styles [36].

Solo and Concerto Datasets. Zhao introduced two complementary corpora in his thesis [35]: the *Solo Dataset* includes 11 hours of studio-recorded monophonic violin audio from 22 performers, partially annotated with onset labels (5 hours). The *Concerto Dataset* consists of 18 hours of polyphonic recordings from nine violinists, with vibrato and partial onset annotations (3924×9 notes). Together they enable studies of both isolated performance and ensemble contexts, supporting tasks such as violinist identification and articulation analysis.

OrchideaSOL (violin subset). OrchideaSOL is a large-scale dataset of extended instrumental techniques for computer-aided orchestration, containing more than 15,000 audio samples across many instruments. The violin subset provides high-quality anechoic studio recordings of extended techniques such as *sul ponticello*, *col legno*, and harmonics. This makes it particularly useful for timbre research focusing on non-standard articulations [38].

MUSC. MUSC is a 34-hour monophonic dataset, curated mainly from online performances, with 22 violinists represented. It offers high-resolution annotations, including frame-level F0 (5.8 ms hop, 10-cent resolution), vibrato, portamento, and note-level transcription aligned to score. The dataset is especially valuable for training and evaluating transcription and expressive-timbre models at fine temporal granularity [34].

URMP. The University of Rochester Multi-Modal Music Performance (URMP) dataset contains 44 multitrack pieces recorded in controlled studio conditions with aligned video. It includes violin, viola, cello, and double bass, allowing for comparative research across the violin family. Rich annotations are provided: score

alignment, F0, onset, and video cues. While not exclusively violin-focused, its multimodal design makes it versatile for tasks combining timbre, score, and performance analysis [39].

RWC Music Database: Musical Instrument Sound (RWC-MDB-I-2001k, violin subset). The RWC Instrument Sound dataset provides 57 monophonic violin recordings, covering single notes at various pitches, dynamic levels (p , mf , f), and playing styles (sustain, staccato, tremolo). Specifically, the violin subset contains $19 \text{ notes} \times 3 \text{ dynamics}$, recorded as single-note files. Beyond violin, the dataset also includes viola, cello, and contrabass subsets with comparable note configurations and metadata (pitch, dynamics, playing style, instrument ID, manufacturer ID), thus supporting violin-family comparative studies [40].

Bach Violin Dataset. The Bach Violin Dataset consists of 6.5 hours of monophonic performances of J. S. Bach’s unaccompanied violin sonatas and partitas (BWV 1001–1006), performed by 17 violinists. Recordings are aligned to the score using Dynamic Time Warping (DTW), ensuring precise note-to-score correspondence. The dataset contains curated recordings, making it a valuable resource for expressive synthesis, performance modelling, and analysis of Baroque repertoire [37].

Chapter 3

Methodology

Overview

DDSP-based Neural synthesis frameworks differ in how they condition the decoder on control parameters.

The **Differentiable Digital Signal Processing (DDSP)** model [7] represents a minimal design: it conditions the decoder on **fundamental frequency (F0)** and **loudness**, with an optional latent $z(t)$ for timbre conditioning. While compact, this configuration is insufficiently informative for timbre control. The latent variable z can capture residual information but is effectively a black-box, reducing interpretability and increasing the risk of overfitting, especially on limited violin data. See Figure 2.

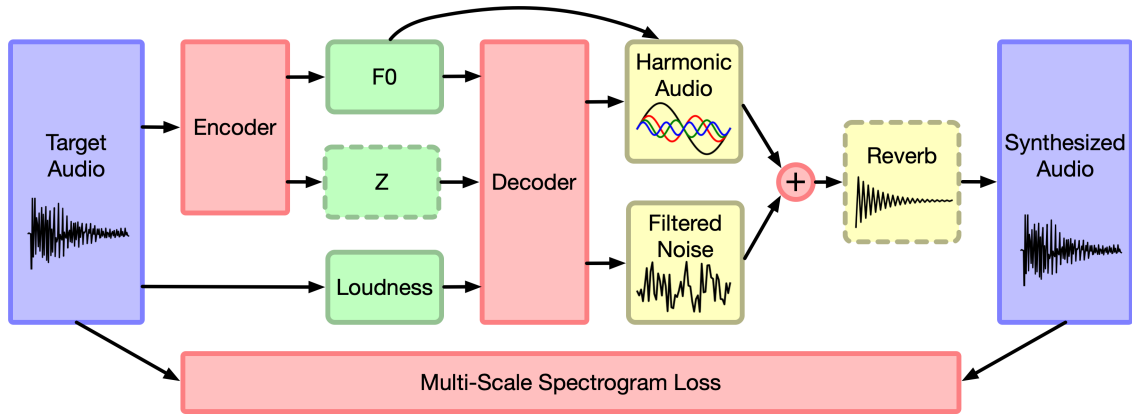


Figure 2: DDSP architecture. Red components are part of the neural network architecture, green components are the latent representation, and yellow components are deterministic synthesizers and effects. Components with dashed borders are not used in all experiments. Namely, z is not used in the model trained on solo violin, and reverb is not used in the models trained on NSynth.

By contrast, the **Glottal-flow LPC Filter with sample-wise LPC (GOLF-ss)** [8] employs **log-F0** and the **mel-spectrogram** as conditioning features. This setup is more informative since the spectrogram carries rich spectral and timbral information. However, it is computationally heavy, prone to noise artefacts, and lacks interpretability, making it cumbersome for lightweight and controllable synthesis (See Figure 3).

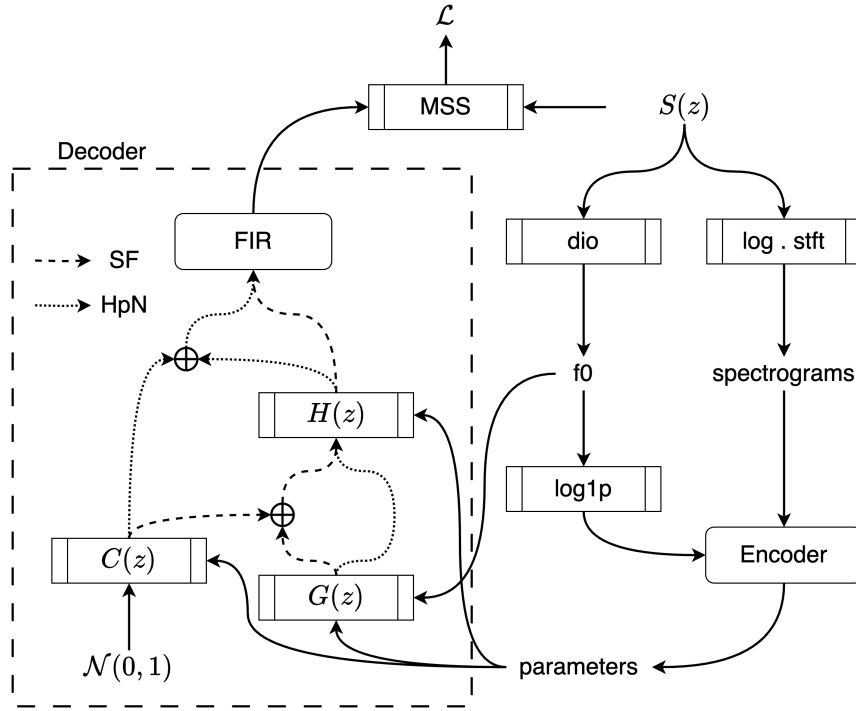


Figure 3: GOLF-ss architecture. $S(z)$ is the speech, $G(z)$ is the harmonic oscillator, $C(z)$ is the noise filter, and $H(z)$ is the vocal tract filter.

Both approaches thus highlight a trade-off: **DDSP** is compact but under-informative, whereas **GOLF-ss** is informative but computationally burdensome and opaque. For violin synthesis, where nuanced timbre control is crucial, neither above is fully satisfactory.

3.0.1 Our Proposal: Timbre-Informed DDSP Violin

We propose **Timbre-Informed DDSP Violin**, a differentiable synthesis framework that extends the original DDSP autoencoder with violin-specific conditioning. Unlike the baseline DDSP conditioned solely on F0 and loudness (optionally a latent z), or GOLF-ss conditioned on F0 and spectrogram features, our model introduces a **compact-and-interpretable conditioning tuple**

$$\mathcal{C}(t) = (f_0(t), e(t), h(t), i(t)),$$

where $f_0(t)$ is the fundamental frequency, $e(t)$ denotes the energy envelope, $h(t)$ is a harmonic-magnitude embedding (hmag embedding, hmag-embed), and $i(t)$ quan-

tifies inharmonicity (inharm). Together, these features provide a lightweight and interpretable alternative to spectrogram-based conditioning.

The architecture (Figure 4) follows the DDSP paradigm of mapping conditioning signals to harmonic and noise synthesisers, while extending the encoder–decoder pipeline with violin-informed features. It is worth mentioning that fundamental-frequency trajectories are obtained using **Constrained Harmonic Resynthesis (CHR)**: an analysis–resynthesis framework applied to the *Violin Etudes* dataset. CHR incorporates violin-specific optimisation strategies to generate clean audio–label pairs for F0, which are then used to finetune a pretrained CREPE network [41]. The finetuned CREPE subsequently provides reliable F0 estimates for the conditioning tuple (see Section 3.2.1).

By conditioning the decoder on this compact tuple, the model disentangles pitch, energy (analogous to loudness in DDSP), and timbre, improving interpretability and synthesis quality while avoiding the burden of dense spectrogram inputs.

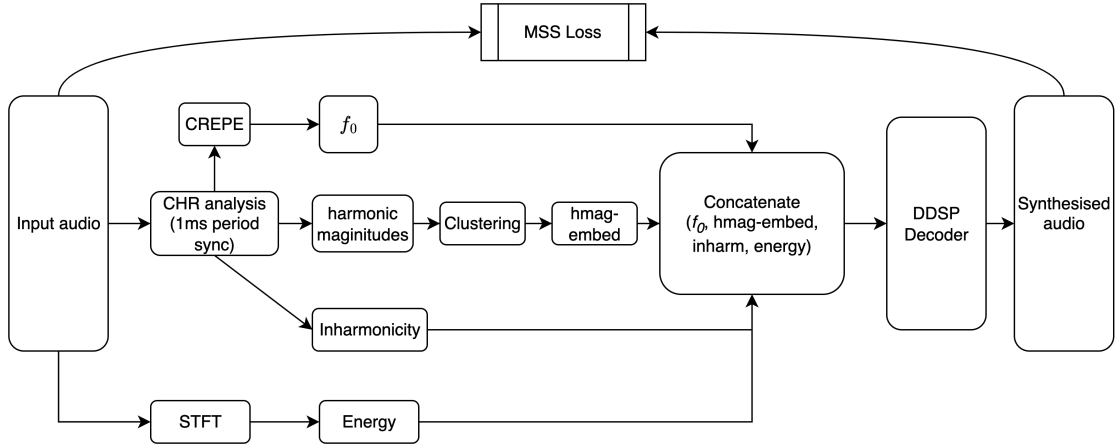


Figure 4: Timbre-Informed DDSP Violin Architecture. The decoder is conditioned on the compact-and-interpretable tuple $\mathcal{C}(t) = (f_0, h, i, e)$, encompassing fundamental frequency, harmonic-magnitude embedding, inharmonicity, and energy. f_0 is estimated using Constrained Harmonic Resynthesis (CHR), applied to the *Violin Etudes* dataset with violin-specific optimisation strategies, producing clean audio–label pairs for finetuning CREPE [41]. The finetuned CREPE then yields reliable f_0 trajectories for synthesis (see Section 3.2.1 for details).

3.1 Timbre Modelling

We model timbre as two **decoupled** elements: (i) a compact **hmag-embed** obtained with **strict** harmonic assignment for stable amplitudes; and (ii) an **inharmonic** measure computed with **relaxed** tolerance to retain true frequency deviations. Section 3.1.1, Section 3.1.2 and Section 3.1.3 describe the protocol and these components respectively.

3.1.1 Decoupling of Harmonic and Inharmonic Components

Spectral Modelling Synthesis (SMS) [42] introduced the harmonic-plus-noise model, which separates a signal into deterministic harmonics and stochastic noise. This principle also underlies DDSF [7], where a harmonic oscillator bank and a noise filter are used as differentiable components. Inspired by this tradition, and by the complexity of violin timbre that combines harmonic structure with deviations such as inharmonicity, we decouple harmonic and inharmonic parts in our framework.

Constrained Harmonic Resynthesis (CHR)¹ [36] builds on SMS-tools² to provide violin-specific refinements for F0 estimation. It applies constraints such as the Instrument-modelling Constraint (IC) and Harmonic Consistency Constraint (HCC), uses CREPE with custom Viterbi decoding, and employs iterative refinement of F0 labels. These adjustments yield more accurate and stable F0 trajectories, which are critical for reliable harmonic analysis.

We adopt the SMS harmonic model to extract harmonic magnitudes (**hmag**) for harmonic analysis, along with harmonic frequencies (**hreq**) for inharmonic analysis. To avoid overlap, different parameter settings are used: for harmonic analysis, CHR is applied with a strict harmonic deviation slope (0.001) to enforce good F0 alignment and yield more consistent hmag across frames, while for inharmonic analysis, a relaxed value (0.02) is chosen after trial-and-error to capture genuine frequency deviations of harmonics. This separation yields two complementary outputs: clean

¹<https://github.com/nctamer/constrained-harmonic-resynthesis>

²<https://github.com/MTG/sms-tools>

hmag for the hmag embedding and genuine hreq with tolerant frequency deviations for the inharmonic measure.

3.1.2 Harmonic Magnitude Embedding

As introduced in Section 3.1.1, harmonic magnitudes (hmag) are extracted with strict harmonic alignment to provide stable amplitude estimates. Although informative, the raw hmag vectors include redundancy and frame-level noise, and thus benefit from being reduced into a compact and interpretable form through clustering and embedding. To obtain a compact and interpretable representation, we transform them through clustering and embedding.

Clustering

Prior studies have shown that clustering can capture latent timbral categories in performance data [12, 13]. Following this idea, we apply k-means clustering to frame-level hmag features extracted at 1 ms resolution. Each frame is assigned to a discrete cluster index, denoted here as c_t , which represents a prototypical harmonic pattern. This step reduces redundancy and groups similar timbral structures into consistent categories.

Embedding

Rather than conditioning the decoder on discrete indices, we embed them into a continuous vector space. Given K clusters and an embedding dimension d , we define a trainable embedding table $E \in \mathbb{R}^{K \times d}$ and obtain the embedding for each frame as

$$\mathbf{e}_t = E[c_t], \quad \mathbf{e}_t \in \mathbb{R}^d.$$

This mapping allows the model to learn relationships between clusters and provides smooth, compact conditioning features. In practice, the embedding is implemented with a standard embedding layer (e.g., `torch.nn.Embedding`).

3.1.3 Inharmonicity Calculation

In an ideal harmonic series (hreq), the n -th partial frequency is located exactly at $n \cdot f_0$. Violin tones, however, often deviate from this relation due to factors such as string stiffness, bowing force, and expressive articulation. To quantify these deviations, we compute an inharmonicity value per frame:

$$I = \frac{2}{f_0 N(N-1)} \sum_{n=2}^N (f_n - n f_0),$$

where f_n is the measured frequency of the n -th harmonic peak, and N is the number of harmonics considered. The frequencies f_n are obtained from harmonic magnitudes extracted with a relaxed harmonic deviation slope (0.02), ensuring that genuine frequency offsets are preserved rather than forced into strict harmonic alignment.

This calculation yields a single scalar per frame that reflects the overall degree of inharmonicity. In our framework, this value is treated as part of the timbre representation and later integrated into the decoder as an additional conditioning parameter (see Section 3.3.2).

3.2 Feature Extraction Pipeline

3.2.1 Constrained Harmonic Resynthesis (CHR)

In Section 3.1, we employed the `harmonicModel` of Constrained Harmonic Resynthesis (CHR) to extract timbre-related **features** such as harmonic magnitude embeddings and inharmonicity. In its original formulation within the *Violin Etudes* dataset [36], CHR was designed as an **iterative F0 labelling framework** to generate reliable pitch trajectories for bowed strings.

The procedure begins with a pre-trainedd CREPE model (**CREPE**₀) with custom Viterbi for fast string jumps while encouraging continuous f0 contours, which provides initial frame-level F0 estimates for audio. These estimates, paired with audio,

are analysed by CHR using SMS harmonic analysis combined with violin-specific constraints: the Instrument-modelling Constraint (IC) and Harmonic Consistency Constraint (HCC). CHR silences inconsistent frames and enforces harmonic alignment, producing clean pairs of (*resynthesised audio*, *constrained F0 labels*).

These pairs are then used to fine-tune CREPE_0 , resulting in a refined estimator (**CREPE-ft**). The new model generates improved F0 trajectories, which are fed back into CHR for another round of constrained analysis and resynthesis. This **iterative refinement loop** continues until convergence, in a process akin to Expectation–Maximisation. Finally, the converged CREPE-ft is applied to the original recordings to produce the **final F0 labels** that enter our conditioning tuple $\mathcal{C}(t)$. The overall pipeline is illustrated in Figure 5.

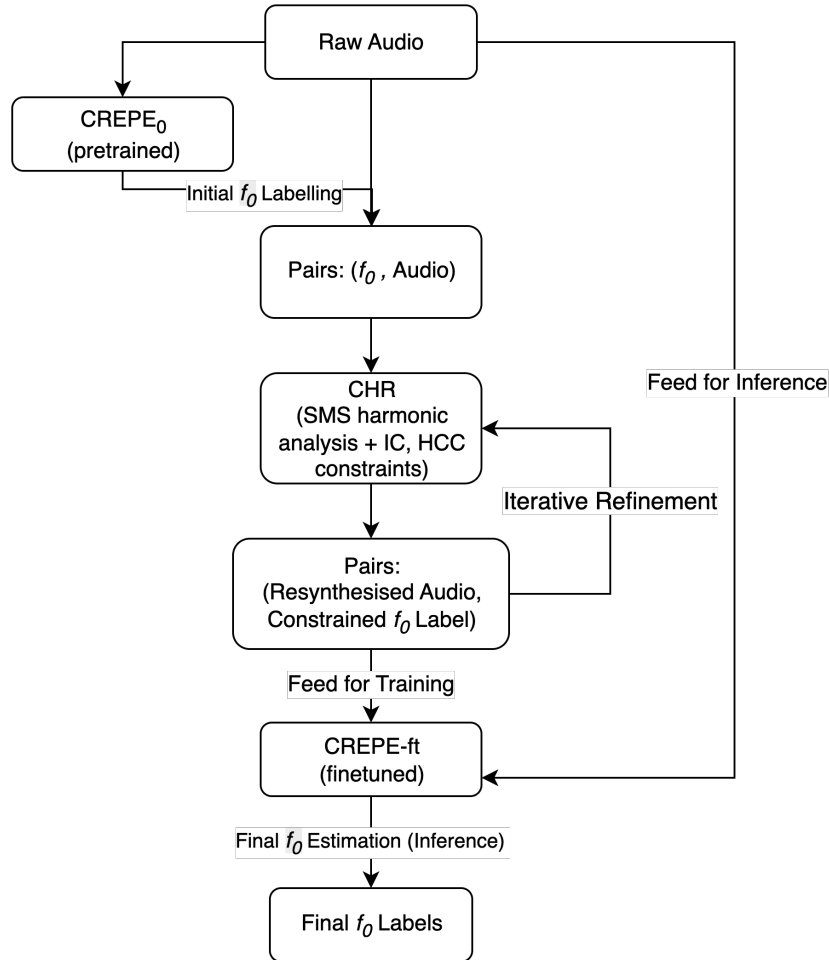


Figure 5: Constrained Harmonic Resynthesis (CHR) pipeline. Starting from a pre-trained CREPE model (CREPE₀), initial F0 estimates are refined by CHR with violin-specific constraints. CHR produces resynthesised audio and constrained F0 labels, which are used to finetune CREPE. The improved predictions are iteratively re-analysed by CHR until convergence. The final finetuned model (CREPE-ft) is then applied to raw audio to obtain reliable F0 trajectories.

3.2.2 Energy Feature

Inspired by **GOLF-ss** [23], we adopt an **energy feature** in place of DDSP’s perceptual loudness, directly derived from the spectral power distribution of the signal. Unlike the loudness curve, which collapses the spectrum into a single perceptual scalar, energy features provide *frequency-aligned, fine-grained information* that is better suited for modelling violin timbre.

The extraction process is as follows:

1. Compute a linear power spectrogram from the STFT of the input waveform:

$$S[f, t] = |X[f, t]|^2.$$
2. For each frame, sample spectral energies at harmonic and half-harmonic frequencies relative to the F0.
3. Split the samples into harmonic energy (integer multiples of F0) and noise energy (DC + half-harmonics).
4. Interpolate both sequences back onto the full frequency axis to obtain a harmonic envelope and a noise envelope.
5. Derive a signal-to-noise (SNR) feature:

$$\text{SNR}(f, t) = \frac{2 \cdot E_{\text{noise}}(f, t)}{E_{\text{harm}}(f, t) + E_{\text{noise}}(f, t) + \epsilon}.$$

Implementation note. The spectrogram is used *only* to derive the energy-related quantities (E_{harm} , E_{noise} , and SNR) and is *not* provided to the model as a conditioning input. Instead, the resulting SNR energy feature $e(t)$ is combined with the other controls to form the compact conditioning tuple

$$\mathcal{C}(t) = (f_0(t), h(t), i(t), e(t)),$$

which directly conditions the decoder.

3.3 Decoder and Conditioning Strategies

3.3.1 DDSP-Based Decoder

Following Engel et al. (2020) [7], our system employs a **DDSP-style decoder** that reconstructs audio by predicting the parameters of three differentiable synthesis components:

- **Harmonic Oscillator Bank:** Generates the pitched component by summing sinusoids at integer multiples of the predicted fundamental frequency f_0 , with amplitudes and phases provided by the decoder.
- **Filtered Noise Synthesizer:** Produces the residual (noisy) component by generating white noise and shaping its spectrum with a learnable filter.
- **Frequency-Domain Reverb:** We explicitly factorise room acoustics into a post-synthesis convolution step using a room impulse response (IR). Convolution is implemented in the frequency domain, reducing computational complexity from $\mathcal{O}(n^3)$ to $\mathcal{O}(n \log n)$ and supporting IRs as long as 10–100k timesteps without slowing training.

This explicit factorisation into harmonic, noise, and reverb components provides strong **physical priors**, resulting in interpretable intermediate representations and controllable synthesis.

3.3.2 Conditioning Configurations

To drive the decoder, we always condition on the fundamental frequency $f_0(t)$ and short-time energy $e(t)$ as basic control signals. We then experiment with adding harmonic and inharmonic descriptors:

Baseline:

$$\mathcal{C}(t) = (f_0(t), e(t))$$

Ablation Variants:

$$\mathcal{C}(t) = (f_0(t), e(t), h(t)) \quad (\text{Add harmonic embedding})$$

$$\mathcal{C}(t) = (f_0(t), e(t), i(t)) \quad (\text{Add inharmonicity})$$

$$\mathcal{C}(t) = (f_0(t), e(t), h(t), i(t)) \quad (\text{Full configuration, proposed})$$

where $h(t)$ is a low-dimensional harmonic-magnitude embedding capturing timbre colour, and $i(t)$ is an inharmonicity measure quantifying deviation from perfect harmonicity.

Summary Table:

Table 2: Conditioning configurations used in our experiments. A check mark ($\checkmark$) indicates that the feature is present.

Configuration	$f_0(t)$	$e(t)$	$h(t)$	$i(t)$
Baseline	$\checkmark$	$\checkmark$		
Add Hmag Embedding	$\checkmark$	$\checkmark$	$\checkmark$	
Add Inharmonicity	$\checkmark$	$\checkmark$		$\checkmark$
Full (Proposed)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

Design Motivation

While mel-spectrogram conditioning provides dense spectral information, it is computationally heavy, noise-sensitive, and less interpretable. Our approach leverages lightweight, feature-based conditioning that captures the most salient perceptual and physical characteristics of violin sound while remaining compact and robust, making it more suitable for controllable synthesis and potential real-time applications.

3.4 Training Objectives

Model training is guided by spectral-domain losses, which constrain both harmonic structures and broadband energy. We explicitly distinguish these training objectives from the evaluation metrics introduced in Chapter 4.4.2.

Multi-Scale Spectral Loss (MSS). Following [7, 43], the primary objective is a multi-scale spectral loss that compares target and reconstructed waveforms in the STFT magnitude domain at multiple resolutions:

$$L_{\text{MSS}}(\hat{x}, x) = \frac{1}{K} \sum_{k=1}^K \left[\alpha \left\| |\text{STFT}_k(\hat{x})| - |\text{STFT}_k(x)| \right\|_1 + (1-\alpha) \left\| \log |\text{STFT}_k(\hat{x})| - \log |\text{STFT}_k(x)| \right\|_1 \right], \quad (3.1)$$

where K denotes the number of STFT configurations, $\alpha \in [0, 1]$ balances linear and logarithmic scales, and $\hat{x}, x$ are the reconstructed and reference audio signals.

Log-F0 Regression Loss (Optional). Following [8], when F0 prediction is enabled (`learn_f0=true`), an auxiliary log-F0 regression loss is applied on voiced frames:

$$L_{f_0} = \frac{1}{|\Omega|} \sum_{t \in \Omega} \left| \log(\hat{f}_0(t) + \varepsilon) - \log(f_0(t) + \varepsilon) \right|, \quad \Omega = \{t : f_0(t) > \tau_0\}, \quad (3.2)$$

where τ_0 is a silence threshold and ε a small stabiliser. This loss supervises the predicted $\hat{f}_0$ against ground-truth f_0 , improving pitch accuracy and reducing pitch-timbre entanglement.

Final Objective. The total training objective is a weighted sum:

$$\mathcal{L} = L_{\text{MSS}} + \lambda_{f_0} L_{f_0},$$

where λ_{f_0} controls the influence of the optional regression loss.

3.5 Summary

This chapter proposed a timbre-informed DDSP framework tailored for violin synthesis. First, we model timbre with two decoupled features: a compact harmonic-magnitude embedding $h(t)$ (strict harmonic assignment) and an inharmonicity measure $i(t)$ (relaxed tolerance). Second, we introduced a compact, interpretable conditioning tuple $\mathcal{C}(t) = (f_0(t), e(t), h(t), i(t))$, where $e(t)$ denotes an energy feature (in

place of DDSP loudness) derived from linear power spectra aligned to $f_0(t)$. Third, we adopted CHR on the *Violin Etudes* corpus to obtain reliable $f_0(t)$ via iterative refinement ($\text{CREPE}_0 \rightarrow \text{CHR} \rightarrow \text{finetune} \rightarrow \text{CREPE-ft}$), which supplies the conditioning pipeline. The decoder follows the DDSP paradigm (harmonic oscillator + filtered noise + reverb), enabling physically interpretable control. Chapter 4 evaluates these design choices through ablations on the conditioning tuple and reports both objective and (planned) subjective results.

Chapter 4

Experiments

Overview

The experiments and code development for **Timbre-Informed DDSP Violin** were carried out on top of the **GOLF-ss repository**¹, which provides a PyTorch Lightning framework, modular DDSP decoders, and ready-to-use evaluation scripts. Building on this foundation, we developed our own implementation, including all code, pretrained checkpoints, and evaluation scripts, which are openly released for reproducibility². Below is the outline of the following sections.

- **4.1 Dataset** – introduces the Violin Etudes dataset, preprocessing, F0 annotation, and data splits.
- **4.2 Preliminary Experiment – DDSP PlayerAcoustics** – tests DDSP’s extendibility by adding a player embedding to capture and transfer acoustic characteristics. Implemented in a dedicated repository³.
- **4.3 Timbre Modelling (Results)** – presents clustering and embedding results from harmonic magnitudes. Code for this section is provided in the `CHR4f0-and-timbre` subfolder of our main repository².

¹<https://github.com/iamycy/golf>

²<https://github.com/kenzhuti/timbre-informed-ddsp-violin>

³<https://github.com/kenzhuti/ddsp-player-acoustics>

- **4.4 Timbre-Informed DDSP Violin** – introduces the experimental version of our proposed architecture, extending DDSP with violin-specific conditioning features. Implementation is available in our main repository².
- **4.5 Results Analysis** – analyses performance across F0 estimators, model selection criteria, subjective listening observations, and identifies the best-performing system.

Together, these experiments examine how compact-and-interpretable violin-specific timbre representations improve DDSP-based synthesis and offer finer control over expressive violin sound generation.

4.1 Dataset

This study employs the **Violin Etudes dataset** [36] for training and evaluation. The collection comprises **226 etudes** performed by **21 violinists**, totalling approximately **27.8 hours** and **925 recordings**. The repertoire was drawn from pedagogical sources such as Wohlfahrt, Kayser, and Paganini, ensuring monophonic violin performances of increasing technical difficulty. These teacher-quality recordings provide a suitable basis for controlled experiments in timbre modelling and synthesis.

Preprocessing. All recordings were originally provided as 48 kHz stereo MP3 files. To obtain a consistent representation, they were converted into 24 kHz mono WAV format using a custom resampling pipeline (`resample_dir.py`). This ensured uniform sampling and eliminated stereo information irrelevant to timbre analysis.

F0 annotation. Two complementary approaches were adopted for pitch annotation:

- **SWIPE-F0** [44]: implemented with a 5 ms hop size (`wav2f0.py`). This configuration, commonly used in singing-voice synthesis, is generally sufficient to capture vocal-level pitch fluctuations.

- **CHR-F0** [36]: following the protocol introduced in the Violin Etudes dataset, Constrained Harmonic Resynthesis was applied at 1 ms resolution. Built upon spectral modelling synthesis [42], this method enforces harmonic consistency and precise periodic alignment, making it particularly effective for capturing fine-grained violin nuances such as vibrato and portamento. Further details are provided in Section 3.2.1

Data split. The dataset was partitioned into **train/validation/test sets** with a ratio of **79% / 14% / 7%**, corresponding to **733 / 130 / 62 recordings** respectively. The split was stratified across performer IDs to preserve player distribution. For violinists with very few recordings, some IDs were excluded from the validation and test sets to avoid unstable partitions. This configuration balances performer diversity with sufficient material for evaluating generalisation in timbre-informed synthesis.

4.2 Preliminary Experiment – DDSP-PlayerAcoustics

Motivation. Motivated by Engel *et al.*’s demonstration of *blind dereverberation* and *transfer of extracted room acoustics* in DDSP [7], and leveraging the shared player labels in the Violin Etudes dataset, we extend the standard conditioning **tuple** $(f(t), l(t), z(t))$ with a **player embedding** p . The goal is to test whether performer-specific acoustic characteristics (room response, microphone colouration) can be disentangled and transferred, rather than capturing playing style (intonation, dynamics, vibrato, etc.).

Method. We use the same training/validation/test splits as described in Section 4.1. Each player ID is mapped to a learned low-dimensional embedding and concatenated to the conditioning tuple at the decoder input:

$$(f(t), l(t), z(t), p) \rightarrow \text{synthesis parameters.}$$

The model is trained end-to-end with the standard multi-resolution spectral loss. No explicit supervision for reverberation or microphone response is used.

Acoustic Transfer Test. At inference, we hold $(f(t), l(t), z(t))$ fixed and switch p to another player ID. The resulting outputs exhibit systematic changes in reverberation decay, spectral envelope tilt, and overall tone colour, consistent with the target player’s acoustics. This indicates that the embedding p learns transferable acoustic information.

Findings.

- Swapping p modifies perceived room characteristics without affecting pitch or loudness contours.
- Spectrograms show consistent decay and envelope changes corresponding to the new player ID.
- The effect remains stable across multiple clips, confirming that acoustic transfer is achieved.

Resources. Code and audio samples are available at <https://github.com/kenzhuti/ddsp-player-acoustics>³ (please cite this repository when using the results).

4.3 Timbre Modelling (Results)

Building on the two-part view in Section 3.1 (harmonic vs inharmonic), this section reports results for the *harmonic* part only: clustering on harmonic magnitudes and the derived *hmag-index* (embedding). The inharmonicity side is deferred to perceptual and objective evaluation in Section 4.5.

Evidence from CHR resynthesis (compact control works). Using the *hmag-index* sequence together with analysis-provided F0 and energy, Constrained Harmonic Resynthesis produces **low-fidelity yet recognisable violin audio resynthesis** via Harmonic-plus-Noise model. This indicates that the compact *hmag-index* retains salient spectral–timbre structure and can function as a lightweight, interpretable control signal.

Clustering behaviour of hmag. Preprocessing with *log compression* and *row-wise normalisation to relative magnitudes* stabilises frame distributions and emphasises spectral *shape* rather than absolute level (Figure 6). We then sweep K and select it using an *elbow plus quality* rule, balancing inertia (elbow), silhouette, and Davies–Bouldin index (Figure 7). A 2D view of the raw hmag already exhibits structure (Figure 9), which clustering discretises into compact, controllable timbre categories. Optional visualisations near the chosen K illustrate the achieved separation (Figure 8, optional).

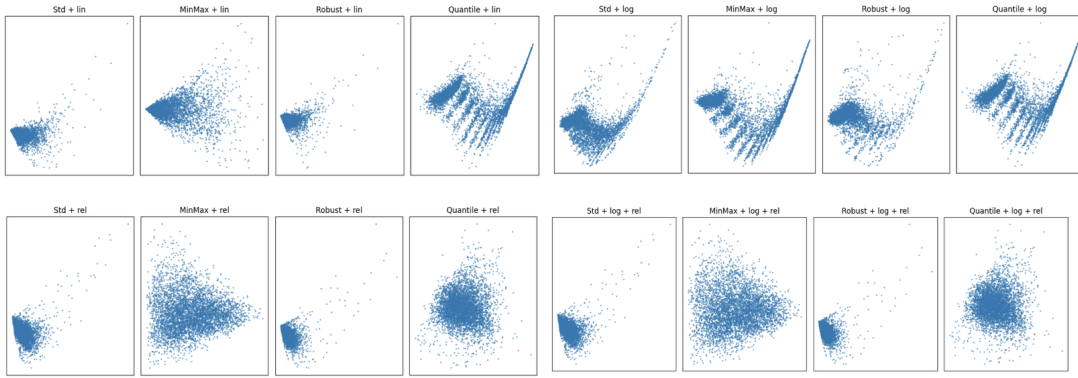


Figure 6: hmag preprocessing: distributions under four scalings and three transforms. Log compression plus row-wise normalisation (relative magnitudes) stabilises heavy-tailed frames and isolates spectral shape from overall level; this configuration is adopted for subsequent clustering.

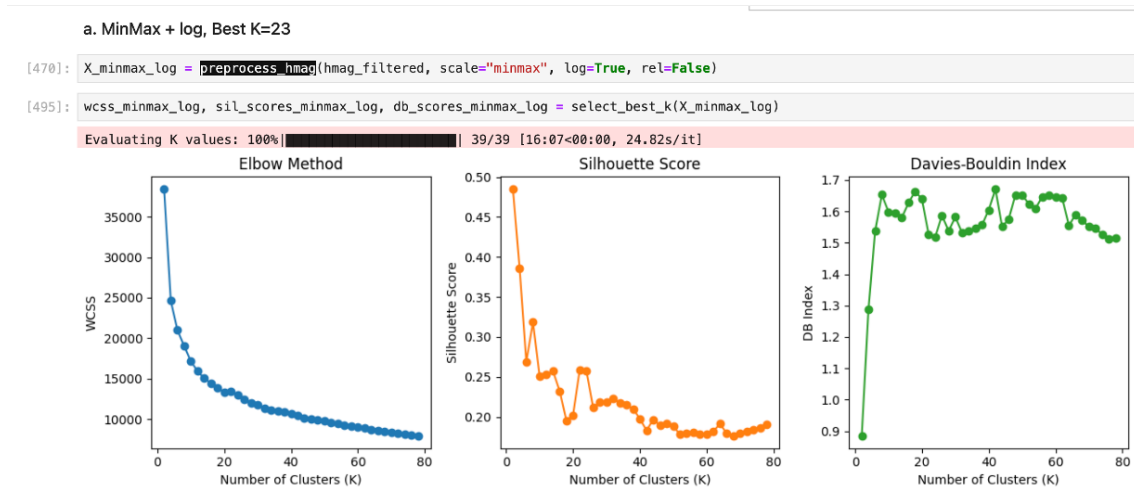


Figure 7: Selecting K (example shown: MinMax plus log). Elbow around the 20–25 range; a local maximum of silhouette and a relatively low Davies–Bouldin index jointly motivate the final choice (in our runs, $K = 23$).

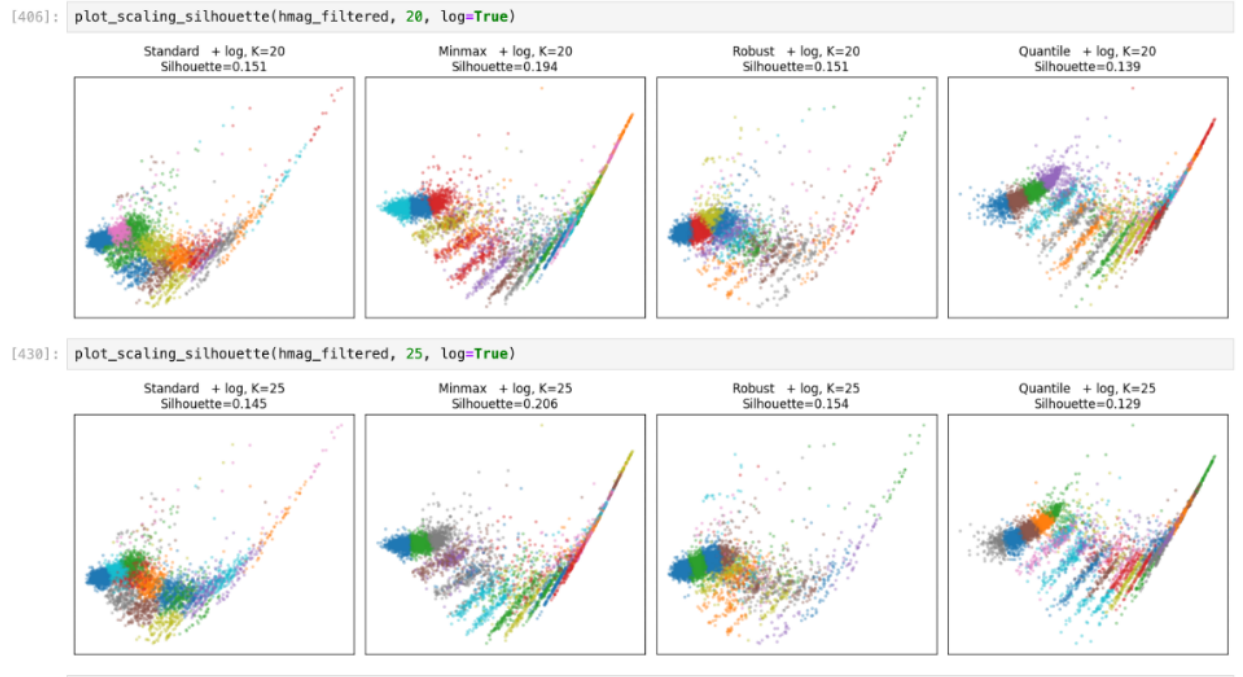


Figure 8: Optional qualitative view of cluster assignments near the chosen K (e.g., $K = 20$ and $K = 25$) under different scalings with log, supporting the selection made from Figure 7.

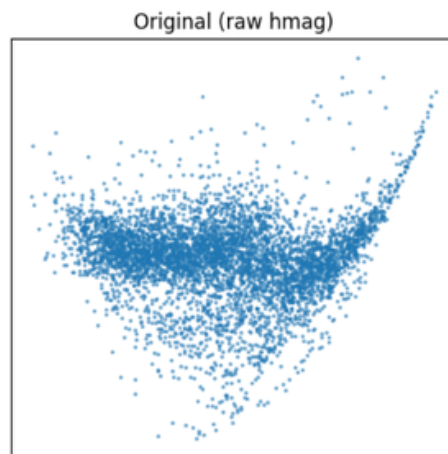


Figure 9: Structure before clustering: a 2D view of raw hmag reveals pre-existing organisation, later discretised into controllable timbre categories.

Link to inharmonicity and next validation step. To validate the complete timbre modelling proposed in Section 3.1, the hmag-index and the inharmonicity features will be *jointly* used as DDSP decoder controls (together with F0 and energy). If this yields improved realism and controllability over baselines, both sides of the timbre factorisation are supported. We report those outcomes in Section 4.5.

4.4 Timbre-Informed DDSP Violin

4.4.1 Experimental Architecture

The methodological architecture proposed in Section 3.0.1 (Figure 4) was designed around a compact and interpretable conditioning tuple:

$$\mathcal{C}(t) = (f_0(t), h(t), i(t), e(t)),$$

consisting of fundamental frequency, harmonic magnitude embeddings, inharmonicity, and energy.

In practice, however, we implemented a hybrid architecture (Figure 10). To improve training stability and to retain high-frequency detail that compact features alone could not fully capture, we concatenated the mel-spectrogram with the compact tuple before feeding it to the DDSP decoder. This hybrid scheme balances interpretability with spectral richness, and provides a more robust learning signal.

Inharmonicity. Unlike F0 and harmonic magnitudes, which are derived from CHR analysis, inharmonicity was computed directly from the STFT. In an ideal harmonic series, partials occur at integer multiples of the fundamental frequency. In violin tones, however, the measured partials often deviate due to bowing force, vibrato, or instrument-specific resonances.

To capture these deviations, we first identify candidate spectral peaks around the expected harmonic bins $\{k_h - 2, \dots, k_h + 2\}$ for each harmonic h . The strongest local maximum is selected, and its position is refined using **three-point parabolic**

interpolation on the triplet (y_-, y_0, y_+) surrounding the detected peak. This procedure yields a sub-bin accurate estimate of the measured harmonic frequency f_h^{meas} .

The frame-level inharmonicity scalar is then computed as:

$$I = \frac{2}{f_0 H(H-1)} \sum_{h=1}^H |f_h^{\text{meas}} - h \cdot f_0|, \quad (4.1)$$

where H is the number of harmonics considered. This descriptor quantifies average harmonic deviations, providing a physically interpretable conditioning signal that complements the discrete timbre embeddings.

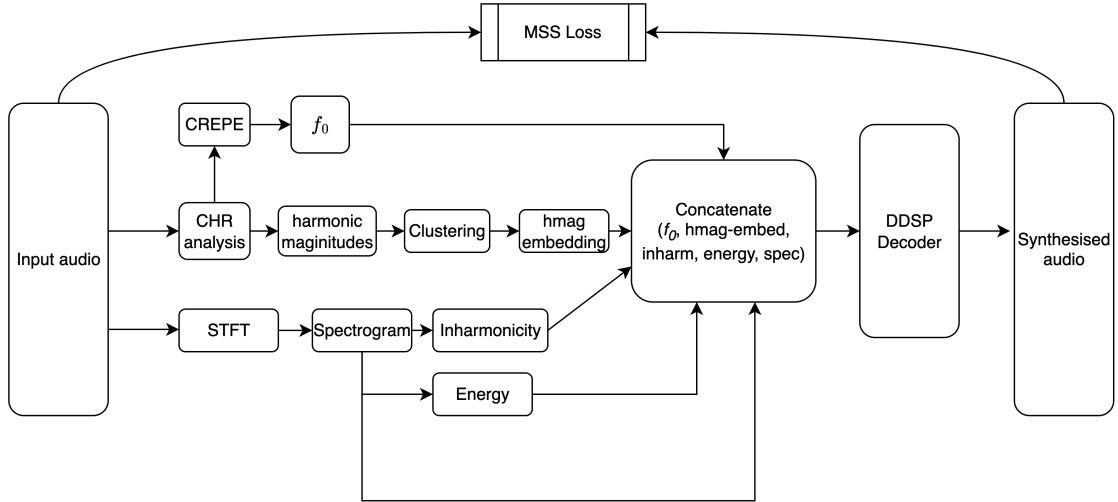


Figure 10: Experimental architecture of Timbre-Informed DDSP Violin. Unlike the methodological design (Figure 4), the experimental setup adopts a hybrid scheme where the mel-spectrogram is concatenated with compact descriptors (f_0, h, i, e) before being fed to the decoder.

4.4.2 Evaluation Metrics

We evaluate our models using both objective and subjective metrics. The objective metrics provide reproducible, quantitative measures, while the subjective metric (MUSHRA) is designed to capture perceptual realism, although it could not be conducted due to time constraints.

Objective Metrics

Mel-Cepstral Distortion (MCD) [45]: Measures the difference between synthesised and reference audio in the Mel-cepstral domain. Lower values indicate greater similarity of spectral envelopes.

Multi-Scale Spectral (MSS) Loss: Assesses reconstruction fidelity across multiple time–frequency resolutions by comparing spectrogram magnitudes. It captures both transient and steady-state accuracy, balancing temporal and spectral fidelity.

Perceptual Evaluation of Speech Quality (PESQ) [46]: Defined in ITU-T P.862, PESQ estimates subjective Mean Opinion Scores (MOS) using auditory perception models. Although designed for speech, it serves as a proxy for perceptual quality in our violin synthesis experiments.

Fréchet Audio Distance (FAD) [47]: Adapted from the Fréchet Inception Distance, FAD compares the distribution of embeddings between real and synthesised audio. Following GOLF-ss [8], we compute FAD using embeddings from the Descript Audio Codec (DAC) [48], a high-fidelity neural audio codec trained on diverse speech, music, and environmental audio. Results are averaged across all test samples.

Subjective Metric

MUltiple Stimuli with Hidden Reference and Anchor (MUSHRA) [49]: A standardised protocol in which listeners rate the perceptual quality of multiple stimuli, including a hidden reference and a degraded anchor, on a 0–100 scale. This test is well suited for evaluating subtle timbral differences in musical audio.

Due to limited time and resources, a formal MUSHRA listening test could not be implemented in this study. Instead, evaluation is restricted to objective metrics, while subjective assessment remains an important direction for future work.

4.4.3 Experimental Configuration

The encoder of the proposed model consists of four layers with channel sizes of 32, 64, 128, and 256. Each layer applies a $4\times$ downsampling, yielding an overall reduction factor of $256\times$. The LSTM module includes three hidden layers, each with 256 units. In the decoder, the harmonic oscillator employs 155 harmonic components, while the noise filter has a frequency resolution of 256. For training, we used a batch size of 128 and a learning rate of $1e-4$, optimised with Adam [50]. Gradient clipping and early stopping were applied to prevent gradient explosion and overfitting, ensuring stable convergence. All experiments were conducted on an NVIDIA A100 GPU (80 GB memory) over 200 epochs.

4.4.4 Experimental Results

Figure 11 from one of our Weights & Biases projects presents the training validation loss curve using F0 estimated by the CHR algorithm with conditioning of both hmag embedding and inharmonicity ⁴, other models show a very similar trend.

⁴<https://wandb.ai/lucein/ddsp-CHRF0-hmag-inharm/workspace?nw=nwuserlucein>



Figure 11: Loss curve of the model trained using CHR F0 with conditioning of hmag embedding and inharmonicity

After training all models, we first evaluate the proposed extensions (additional hmag embedding and/or inharmonicity) to the DDSP baseline on the full Violin Etudes dataset, using two different fundamental frequency (F0) estimation methods (SWIPE and CHR). This allows us to compare the impact of the hmag embedding and the inharmonicity feature under different F0 estimation strategies.

Table 3 (SWIPE-F0)

Table 3 reports the results obtained with SWIPE F0. The baseline DDSP model is compared with three variants: DDSP-hmag (adding the hmag embedding), DDSP-inharm (adding the inharmonicity feature), and DDSP-hmag-inharm (combining both).

Table 3: Experimental results on the full dataset using SWIPE-f0. ↓: lower is better; ↑: higher is better.

Model	avg_mcd↓	avg_mss_loss↓	PESQ↑	FAD↓
Notes: * = best in this table; ** = best across all tables.				
DDSP	4.2973	3.7848	1.3127	260.4146**
DDSP-hmag	4.2308	3.7714	1.2548	277.9285
DDSP-inharm	4.2852	3.7834	1.3193*	288.4417
DDSP-hmag-inharm (BEST)	4.2129*	3.7713*	1.3177	284.5082

Table 4 (CHR-F0)

Table 4 shows the corresponding results when using CHR F0 on the same dataset. The same model variants are evaluated, allowing us to verify the consistency of the improvements across a different F0 estimation method.

Table 4: Experimental results on the full dataset using CHR-f0. ↓: lower is better; ↑: higher is better.

Model	avg_mcd↓	avg_mss_loss↓	PESQ↑	FAD↓
Notes: * = best in this table; ** = best across all tables.				
DDSP	4.2839	3.8431	1.3144	279.1999
DDSP-hmag	4.2740	3.8112	1.3172	274.3401
DDSP-inharm	4.2534	3.8021	1.3252*	271.6798
DDSP-hmag-inharm (BEST across all tables)	4.2391*	3.7825*	1.3220	269.1621*

Table 5 (CHR-F0 player01)

To reduce the influence of inter-player differences in instrument and recording acoustics across the 21 violinists, we also conducted a single-subject experiment using only the recordings of player 01, who has the largest number of samples (260 out of 925). In this controlled setting, we trained and evaluated models with CHR F0 to test whether the hmag embedding and the inharmonicity feature still provide improvements when inter-player variability is minimised. Table 5 reports the results on this subset, comparing a re-trained DDSP baseline with the DDSP-hmag-inharm model.

Table 5: Experimental results on the Solo-player01 dataset (CHR-f0). ↓: lower is better; ↑: higher is better.

Model	avg_mcd↓	avg_mss_loss↓	PESQ↑	FAD↓
Notes: * = best in this table; ** = best across all tables.				
DDSP-bsl	4.1616	3.6356	1.3344	346.9540*
DDSP-hmag-inharm (BEST)	4.1468***	3.6220***	1.3410***	349.6216

To further evaluate the proposed model, we synthesised several violin pieces. Representative audio samples are available online⁵, as illustrated in Figure 12.

Audio Demos for Timbre-Informed DDSP Violin

Audio demos trained on the [Violin Etudes dataset](#).

Three tables compare different models with different f0 estimators: CHR (Constrained Harmonic Resynthesis) only for player 01, and CHR vs SWIPE (A Sawtooth Waveform Inspired Pitch Estimator for Speech and Music) for all 21 players.

CHRf0-Only-Player01

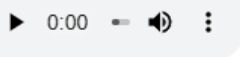
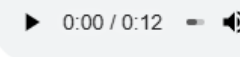
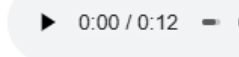
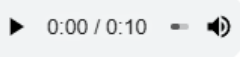
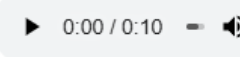
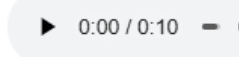
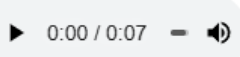
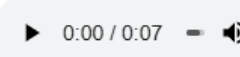
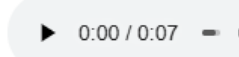
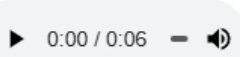
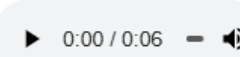
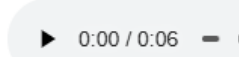
Test clip	Reference	DDSP-bsl	DDSP-hmag-inharm
player01-1			
player01-2			
player01-3			
player01-4			

Figure 12: Webpage interface providing access to synthesised violin audio demonstrations.

⁵<https://kenzhuti.github.io/audio-demo/>

4.5 Results Analysis

The evaluation presented in Section 4.4 directly relates to the central aim of this thesis: to design a timbre-informed neural violin synthesiser that extends DDSP with hmag embeddings and inharmonicity. Building on the Violin Etudes dataset [36], which provides high-precision F0 labels and diverse player recordings, our experiments tested whether these extensions improve synthesis quality and performance metrics under different F0 estimation strategies (SWIPE and CHR).

4.5.1 SWIPE-F0 vs. CHR-F0

The evaluation in Section 4.4 highlights performance differences under SWIPE-F0 and CHR-F0 estimation. Both SWIPE-F0 models and CHR-F0 models improve with additional conditioning of hmag embeddings and inharmonicity, aligning with the thesis goal of enhancing DDSP for violin synthesis using hmag embeddings and inharmonicity. However, the models using CHR-F0 are generally better than the models using SWIPE-F0.

While SWIPE-F0 results already show performance gains, an anomaly was observed: the baseline DDSP model achieved the best FAD score under SWIPE-F0 (260.4146). A likely explanation is the interaction between temporal resolution and the DAC-based FAD metric: SWIPE-F0 uses a 5 ms hop that aligns more closely with DAC’s 10 ms frame rate, whereas timbre-informed models rely on 1 ms features. The baseline therefore produces smoother, speech-like statistics that the speech-trained codec favours, yielding the lowest FAD score (260.4146) despite lacking violin-specific detail.

In contrast, CHR F0 estimation leverages spectral modelling synthesis, harmonic consistency constraints, and instrument-informed priors, capturing violin periodicity and harmonic phase continuity more accurately. This results in significant improvements not only in MCD and MSS, but also in FAD (a relative reduction of 28.1% compared to baseline). These findings align with previous work on Violin Etudes, which reports that CHR-F0 yields higher temporal precision and better

handles fast string crossings than conventional estimators. Given these results, we focus subsequent comparisons on CHR-F0 models, as they better match the design objectives of this work and provide more reliable evaluations for violin synthesis.

4.5.2 Model Selection for Comparisons

Among all variants, our proposed DDSP-hmag-inharm consistently outperforms other models in MCD and MSS, while DDSP-inharm shows a marginal advantage in PESQ (≤ 0.0016). Because the combined model is superior in most metrics and shows more balanced performance, it is selected as the primary model for comparison against the baseline in the subsequent evaluation and experiments. This avoids redundancy and highlights the contribution of integrating both hmag embedding and inharmonicity features.

4.5.3 Subjective Observations and Player01 Controlled Models

Subjective listening proves the objective results: DDSP-hmag-inharm reduces the “tube-like” resonance and background noise observed in baseline models, particularly in low-frequency regions. However, experiments on the Solo-player01 subset (260 samples) revealed relatively poorer FAD, likely due to the smaller amount of training data. Even so, the improvements in MCD, MSS, and PESQ, together with perceptual enhancements, confirm that different player acoustics introduce distribution shifts that the model partially learns. This motivates future work on incorporating player-ID embeddings or curating anechoic datasets to better disentangle instrument timbre from recording and instrument acoustics.

4.5.4 Overall Best Model

Considering all three tables jointly, **DDSP-hmag-inharm (CHR-F0)** emerges as the most balanced model:

- **avg MCD $\downarrow$ = 4.1468**

- $\text{avg MSS}\downarrow = 3.6220$
- $\text{PESQ}\uparrow = 1.3410$

Although the baseline achieved the best FAD under SWIPE-F0, the CHR-F0 variant of DDSP-hmag-inharm yields stronger FAD improvements relative to its baseline and the best other metrics in the table, which makes it the most appropriate configuration for subsequent analysis.

4.5.5 Summary

In summary, the results indicate four main findings:

1. CHR F0 estimation is more robust than SWIPE for violin, improving both spectral and perceptual measures.
2. Integrating hmag embedding and inharmonicity leads to systematic improvements over the DDSP baseline.
3. The fully configured DDSP-hmag-inharm model offers the best performance under both F0 estimation strategies.
4. Variability in recording acoustics was found to introduce confounding noise into the model, negatively affecting synthesis quality (as observed in Section 4.5.3).

The broader implications of these results are addressed in Chapter 5.

Chapter 5

Discussion

5.1 Contributions and Findings

Methodological — a compact, interpretable conditioning tuple integrating violin timbre. We propose a lightweight control set $\mathcal{C}(t) = (f_0(t), e(t), h(t), i(t))$, where $h(t)$ is a harmonic-magnitude embedding and $i(t)$ an explicit inharmonicity feature. Together, they capture complementary aspects of violin timbre (harmonic structure and departures from harmonicity). F0 trajectories are estimated through CHR-based finetuned CREPE introduced in Violin Etudes paper. This preserves DDSP’s factorised interpretability while avoiding spectrogram-heavy inputs.

Experimental — effectiveness of the timbre representation. Ablations show consistent gains once $h(t)$ and $i(t)$ are introduced. Under CHR-F0, *DDSP-hmag-inharm* is the most balanced across spectral and perceptual metrics; under SWIPE-F0, improvements persist although FAD exhibits a known bias. Overall, the tuple (f_0, e, h, i) offers a compact and controllable route for timbre-informed conditioning in DDSP.

Reproducibility and resources. We release code, trained checkpoints, and audio demos in the main repository; a preliminary companion study suggests extensibility to player-related acoustic factors, without altering F0 or energy controls.¹

¹Main repo: github.com/kenzhuti/timbre-informed-ddsp-violin; companion repo:

5.2 Limitations

Lack of subjective evaluation. This study reports objective metrics only; controlled listening tests (e.g., MUSHRA) were not conducted and remain future work.

Hybrid conditioning in the implemented system. Our experimental architecture concatenates a mel-spectrogram with the compact tuple for stability. This partly dilutes the “compact-only” claim; a spectrogram-free variant based solely on (f_0, e, h, i) has not yet been validated.

Metric domain bias. FAD is computed using embeddings from a codec primarily trained on speech, which may bias scores and under-represent music-specific aspects; however, music-trained embeddings were not evaluated.

Limited control coverage. The current decoder does not explicitly model transients or note transitions, and uses a generic excitation and resonator design; explicit bow-string excitation and a violin body resonance filter are deferred to future work.

Dataset scope and generalisation. Results are reported on a single dataset with various recording and instrument conditions; cross-dataset and cross-condition generalisation (players, rooms, instruments) was not systematically assessed.

5.3 Future Work

Subjective listening tests. We will complement objective metrics with controlled listening experiments (e.g., MUSHRA) on the test set. This includes clear instructions, hidden references, anchors, and sufficient listeners for statistical analysis. We will study how perceived quality relates to spectral fidelity and to the proposed conditioning tuple.

Compact-only conditioning. In our *experimental architecture*—the practical, implemented setup—we used a hybrid scheme that concatenates a mel-spectrogram with the compact tuple for training stability and convergence. By contrast, our

github.com/kenzhuti/ddsp-player-acoustics.

methodological architecture targets a spectrogram-free variant that relies solely on (f_0, e, h, i) . We will conduct systematic ablations to quantify the trade-off between compactness, interpretability, and reconstruction quality, and to verify that the compact-only configuration can replace the hybrid in practice.

Physical excitation and violin body resonance. We will replace glottal-flow wavetables with a bow-string excitation module, and add a learnable violin body resonance filter based on linear prediction. The source will approximate the bowing mechanism (alternating stick-slip), while the filter will act as a violin-specific LPC resonator. We will compare frame-wise LPC (time-invariant within a frame) with differentiable time-varying LP (sample-wise coefficients), and evaluate simple, stable implementations. We will keep a DDSP-style factorisation into source, resonator, noise, and reverb to maintain interpretability and control [7, 23, 8, 42].

Modelling transients and note transitions. To improve realism around onsets, offsets, and bow changes, we will add a transient pathway or attack-aware losses, following the idea of complementing steady-state spectral modelling with explicit transient/transition modelling [51, 52].

Other interpretable timbre representations for conditioning. Beyond harmonic-magnitude embeddings and inharmonicity, we will test alternative learned features as conditioning signals. Candidates include deep clustering (HuBERT-like) [53], neural codecs such as LSCodec [18], CoDiCodec [54], and DAC [48], and disentangled timbre representations from pitch [14, 15, 16]. We will compare their dimensionality, latency, controllability, and correlations with listening scores.

Extending Control to Player-related Acoustics. We plan to integrate the player embedding from DDSP-PlayerAcoustics into Timbre-Informed DDSP Violin to control player-related acoustics (e.g., room response, microphone colouration, and instrument body resonances) independently of timbre and pitch.

High-fidelity violin corpus. We plan to curate a studio-grade dataset with consistent microphone setups, calibrated dynamics, and anechoic or low-reverb conditions. The corpus will include taxonomies of articulations and note transitions, enabling

both supervised learning of controls and fair evaluation across playing conditions.

Architecture variants. We will explore diffusion and Transformer-based decoders under the same conditioning and datasets, using the DDSP baseline as the low-latency, interpretable reference (e.g., diffusion-based violin synthesis [32]).

Evaluation with music-trained embeddings. Current FAD scores rely on DAC embeddings, which are trained primarily on speech and may bias results. As future work, we will explore alternatives such as the Jukebox encoder [55], trained on large-scale music recordings, and MuLan [56], optimised for music retrieval and tagging. These music-trained embeddings are better suited to capture musical structure and timbre, and could provide more reliable evaluation for violin synthesis.

Chapter 6

Conclusions

This thesis investigated timbre-informed neural violin synthesis. We adopted a DDSP-based decoder and proposed a compact-and-interpretable conditioning tuple $\mathcal{C}(t) = (f_0, e, h, i)$, where e is SNR energy feature, h denotes a harmonic-magnitude embedding and i an inharmonicity feature. In the implemented system, externally estimated F0 (CHR and SWIPE) was used for conditioning; the focus of the work is on how timbre information is represented and injected into the synthesiser.

The work yields three main outcomes. First, it shows that violin timbre can be expressed effectively with two complementary facets—harmonic structure and departures from harmonicity—and injected as lightweight controls while preserving DDSP’s factorised design. Second, ablations indicate consistent improvements over a baseline DDSP once the timbre features are introduced; the *DDSP-hmag-inharm* configuration with CHR–F0 provides the most balanced performance across spectral and perceptual metrics (see Section 4.4). Third, code, trained checkpoints, and audio demonstrations are released to support reproducibility and follow-up research.

Relative to the stated objectives (Section 1.2), the proposed timbre modelling is compact and effective; the conditioning design is interpretable and practical (with a hybrid implemented system that concatenates a mel-spectrogram for stability), and the synthesis quality improves against the baseline. Remaining limitations include the lack of controlled subjective listening tests, potential metric-domain bias, the

integration of heavy spectrogram into conditioning tuple in the implemented system, and the current dataset scope (see Section 5.2).

These findings suggest a practical route to controllable, interpretable violin synthesis that balances compactness with sound quality. The framework is readily extensible to additional factors—including player, room, and instrument acoustics—and to alternative timbre representations. Future work will prioritise controlled listening tests (e.g., MUSHRA), training a compact-only variant without spectrograms, adding a bow–string excitation with a violin body resonance filter, and improving the modelling of transients and note transitions (see Section 5.3).

In summary, compact and interpretable timbre control enhances both quality and controllability in violin synthesis, providing a concrete foundation for perceptual evaluation and physically informed extensions.

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Bibliography

- [1] McAdams, S. Timbre space: Timbre as a structuring force in music. In Siedenburg, K., Saitis, C., McAdams, S., Popper, A. N. & Fay, R. R. (eds.) *Timbre: Acoustics, Perception, and Cognition*, vol. 69 of *Springer Handbook of Auditory Research*, 211–243 (Springer International Publishing, Cham, 2019).
- [2] Saitis, C. & Weinzierl, S. The semantics of timbre. In Siedenburg, K., Saitis, C., McAdams, S., Popper, A. N. & Fay, R. R. (eds.) *Timbre: Acoustics, Perception, and Cognition*, vol. 69 of *Springer Handbook of Auditory Research*, 119–149 (Springer International Publishing, Cham, 2019).
- [3] Woodhouse, J. The acoustics of the violin: a review. *Reports on Progress in Physics* **77**, 115901 (2014).
- [4] Hung, T.-Y. *et al.* A study on synthesizing expressive violin performances: Approaches and comparisons. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)* (San Francisco, USA, 2024).
- [5] Nizami, M. & Lestari, D. P. A dt-neural parametric violin synthesizer. In *2021 International Conference on Electrical Engineering and Informatics (ICEEI)*, 1–6 (2021).
- [6] Jonason, N. *The Control-Synthesis Approach for Making Expressive and Controllable Neural Music Synthesizers*. Master’s thesis, KTH Royal Institute of Technology (2020). URL <https://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-287553>. Media Technology and Computer Science and Engineering, Second Cycle, 30 Credits.

- [7] Engel, J., Hantrakul, L., Gu, C. & Roberts, A. Ddsp: Differentiable digital signal processing. In *International Conference on Learning Representations (ICLR)* (2020). URL <https://openreview.net/forum?id=B1x1ma4tDr>.
- [8] Yu, C.-Y. & Fazekas, G. Differentiable time-varying linear prediction in the context of end-to-end analysis-by-synthesis. In *Interspeech 2024*, 1820–1824 (ISCA, 2024). URL <http://dx.doi.org/10.21437/Interspeech.2024-1187>. Interspeech_2024.
- [9] Brown, J. C. Calculation of a constant q spectral transform. *Journal of the Acoustical Society of America* **89**, 425–434 (1991).
- [10] Grossmann, A. & Morlet, J. Decomposition of hardy functions into square integrable wavelets of constant shape. *SIAM Journal on Mathematical Analysis* **15**, 723–736 (1984).
- [11] Davis, S. B. & Mermelstein, P. Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. *IEEE Transactions on Acoustics, Speech, and Signal Processing* **28**, 357–366 (1980).
- [12] Perez, A. & Ramirez, R. Towards realistic and natural synthesis of musical performances: Performer, instrument and sound modeling. In *Proceedings of the Third Vienna Talk on Music Acoustics* (2015). URL https://viennatalk2015.mdw.ac.at/proceedings/ViennaTalk2015_submission_94.pdf.
- [13] Mannone, M., Distefano, V. & Santini, G. Classes of colors and timbres: A clustering approach. *Electronic Journal of Applied Statistical Analysis* **15**, 588–605 (2022).
- [14] Luo, Y., Agres, K. & Herremans, D. Learning disentangled representations of timbre and pitch for musical instrument sounds using gaussian mixture variational autoencoders (2019). 1906.08152.
- [15] Luo, Y. *et al.* Disentangling multi-instrument music audio for source-level pitch and timbre manipulation. In *Audio Imagination: NeurIPS 2024 Workshop*

- on AI-Driven Speech, Music, and Sound Generation* (2024). URL <https://openreview.net/forum?id=Zhc6ZFmVM0>.
- [16] Tanaka, K., Nishikimi, R., Bando, Y., Yoshii, K. & Morishima, S. Pitch-timbre disentanglement of musical instrument sounds based on vae-based metric learning. In *ICASSP 2021 - IEEE International Conference on Acoustics, Speech and Signal Processing*, 111–115 (2021).
- [17] Nakashima, F., Nakamura, T., Takamune, N., Fukayama, S. & Saruwatari, H. Hyperbolic timbre embedding for musical instrument sound synthesis based on variational autoencoders. In *APSIPA ASC 2022 - Asia-Pacific Signal and Information Processing Association Annual Summit and Conference*, 735–742 (2022).
- [18] Guo, Y. *et al.* Lscodec: Low-bitrate and speaker-decoupled discrete speech codec (2025). 2410.15764.
- [19] Hayes, B., Shier, J., Fazekas, G., McPherson, A. & Saitis, C. A review of differentiable digital signal processing for music and speech synthesis. *Frontiers in Signal Processing* **3**, 1284100 (2024).
- [20] Schwarz, D. Corpus-based concatenative synthesis. *IEEE Signal Processing Magazine* 92–104 (2007).
- [21] Bilbao, S. *Numerical Sound Synthesis: Finite Difference Schemes and Simulation in Musical Acoustics* (John Wiley & Sons, 2009).
- [22] Yeh, Y.-T. *et al.* Ddsp guitar amp: Interpretable guitar amplifier modeling. In *ICASSP* (2025).
- [23] Yu, C.-Y. & Fazekas, G. Singing voice synthesis using differentiable lpc and glottal-flow-inspired wavetables. In *ISMIR* (2023).
- [24] Liu, Y., Jin, C. & Gunawan, D. Ddsp-sfx: Acoustically-guided sound effects generation with differentiable digital signal processing. In *DAFx* (2024).

- [25] Gutiérrez, E., Font, F., Serra, X. & Wyse, L. A statistics-driven differentiable approach for sound texture synthesis and analysis. In *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)* (Ancona, Italy, 2025).
- [26] Renault, L., Mignot, R. & Roebel, A. Ddsp-piano: A neural sound synthesizer informed by instrument knowledge. *Journal of the Audio Engineering Society* **71**, 552–565 (2023).
- [27] Li, S. *et al.* Snakegan: A universal vocoder leveraging ddsp prior knowledge and periodic inductive bias. In *ICME*, 1703–1708 (2023).
- [28] Wu, Y. *et al.* Midi-ddsp: Detailed control of musical performance via hierarchical modeling. In *ICLR* (2022).
- [29] Caspe, F., McPherson, A. & Sandler, M. Ddx7: Differentiable fm synthesis of musical instrument sounds. In *ISMIR* (2022).
- [30] Kawamura, M. *et al.* Differentiable digital signal processing mixture model for synthesis parameter extraction from mixture of harmonic sounds. In *ICASSP*, 941–945 (2022).
- [31] Shan, S., Hantrakul, L., Chen, J., Avent, M. & Trevelyan, D. Differentiable wavetable synthesis. In *ICASSP*, 4598–4602 (2022).
- [32] Kim, D., Dong, H.-W. & Jeong, D. Violindiff: Enhancing expressive violin synthesis with pitch bend conditioning. In *ICASSP 2025 - IEEE International Conference on Acoustics, Speech and Signal Processing*, 1–5 (2025).
- [33] Volhejn, V. *Accelerating Neural Audio Synthesis*. Master’s thesis, ETH Zürich (2022). URL <http://hdl.handle.net/20.500.11850/571861>. Accessed: Sept. 13, 2025.
- [34] Tamer, N. C., Özer, Y., Müller, M. & Serra, X. MUSC: High-Resolution Violin Transcription Using Weak Labels. In *Proceedings of the 24th International Society for Music Information Retrieval Conference (ISMIR)* (2023).

- [35] Zhao, Y. *A Study in Violinist Identification using Short-term Note Features*. Ph.D. thesis, Queen Mary University of London (2022). URL <https://qmro.qmul.ac.uk/xmlui/handle/123456789/82067>.
- [36] Tamer, N. C., Ramoneda, P. & Serra, X. Violin etudes: A comprehensive dataset for f0 estimation and performance analysis. In *Proceedings of the 23rd International Society for Music Information Retrieval Conference (ISMIR)*, 616–623 (2022).
- [37] Dong, H.-W., Zhou, C., Berg-Kirkpatrick, T. & McAuley, J. Deep performer: Score-to-audio music performance synthesis. In *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 951–955 (2022).
- [38] Cella, C. E. *et al.* Orchideasol: a dataset of extended instrumental techniques for computer-aided orchestration (2020). URL <https://arxiv.org/abs/2007.00763>. 2007.00763.
- [39] Li, B., Liu, X., Dinesh, K., Duan, Z. & Sharma, G. Urmp- creating a multitrack classical music performance dataset for multimodal music analysis: Challenges, insights, and applications. In *IEEE Transactions on Multimedia*, vol. 21, 522–535 (2019).
- [40] Goto, M., Hashiguchi, H., Nishimura, T. & Oka, R. Rwc music database: Popular, classical and jazz music databases. In *Proceedings of the 3rd International Conference on Music Information Retrieval (ISMIR)* (2002).
- [41] Kim, J. W., Salamon, J., Li, P. & Bello, J. P. CREPE: A convolutional representation for pitch estimation. In *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 161–165 (IEEE, Calgary, AB, Canada, 2018).
- [42] Serra, X. & Smith, J. O. Spectral modeling synthesis: A sound analysis/synthesis system based on a deterministic plus stochastic decomposition. *Computer Music Journal* **14**, 12–24 (1990).

- [43] Schwär, S. & Müller, M. Multi-scale spectral loss revisited. *IEEE Signal Processing Letters* **30**, 1712–1716 (2023).
- [44] Camacho, A. & Harris, J. G. A sawtooth waveform inspired pitch estimator for speech and music. *The Journal of the Acoustical Society of America* **124**, 1638–1652 (2008).
- [45] Kubichek, R. Mel-cepstral distance measure for objective speech quality assessment. In *Proceedings of IEEE Pacific Rim Conference on Communications, Computers and Signal Processing*, vol. 1, 125–128 (IEEE, 1993).
- [46] Rix, A. W., Beerends, J. G., Hollier, M. P. & Hekstra, A. P. Perceptual evaluation of speech quality (pesq) – a new method for speech quality assessment of telephone networks and codecs. In *2001 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, vol. 2, 749–752 (IEEE, 2001).
- [47] Kilgour, K., Zuluaga, M., Roblek, D. & Sharifi, M. Fréchet audio distance: A metric for evaluating music enhancement algorithms. *arXiv preprint arXiv:1812.08466* (2018).
- [48] Kumar, R., Seetharaman, P., Luebs, A., Kumar, I. & Kumar, K. High-fidelity audio compression with improved rvqgan. *arXiv preprint arXiv:2306.06546* (2023). URL <https://arxiv.org/abs/2306.06546>.
- [49] International Telecommunication Union. Method for the subjective assessment of intermediate quality level of audio systems (mushra). Tech. Rep. Recommendation ITU-R BS.1534-3, ITU-R, Geneva, Switzerland (2015). <https://www.itu.int/rec/R-REC-BS.1534/en>.
- [50] Kingma, D. P. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [51] Verma, T. S. & Meng, T. H. Y. Extending spectral modeling synthesis with transient modeling synthesis. *Computer Music Journal* **24**, 47–59 (2000).

- [52] Yang, L., Chew, E. & Rajab, K. Z. Logistic modeling of note transitions. In Collins, T., Meredith, D. & Volk, A. (eds.) *Mathematics and Computation in Music (MCM)*, vol. 9110 of *Lecture Notes in Computer Science*, 161–172 (Springer International Publishing, Cham, 2015).
- [53] Hsu, W.-N. *et al.* HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (2021).
- [54] Pasini, M., Lattner, S. & Fazekas, G. Codicodec: Unifying continuous and discrete compressed representations of audio. *arXiv preprint arXiv:2509.09836* (2025).
- [55] Dhariwal, P. *et al.* Jukebox: A generative model for music. In *Proceedings of the International Conference on Learning Representations (ICLR)* (2020). URL <https://arxiv.org/abs/2005.00341>.
- [56] Huang, Q., Kim, J. W., Salamon, J. & Bello, J. P. Mulan: A joint embedding of music audio and natural language. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)* (2022). URL <https://arxiv.org/abs/2208.12415>.